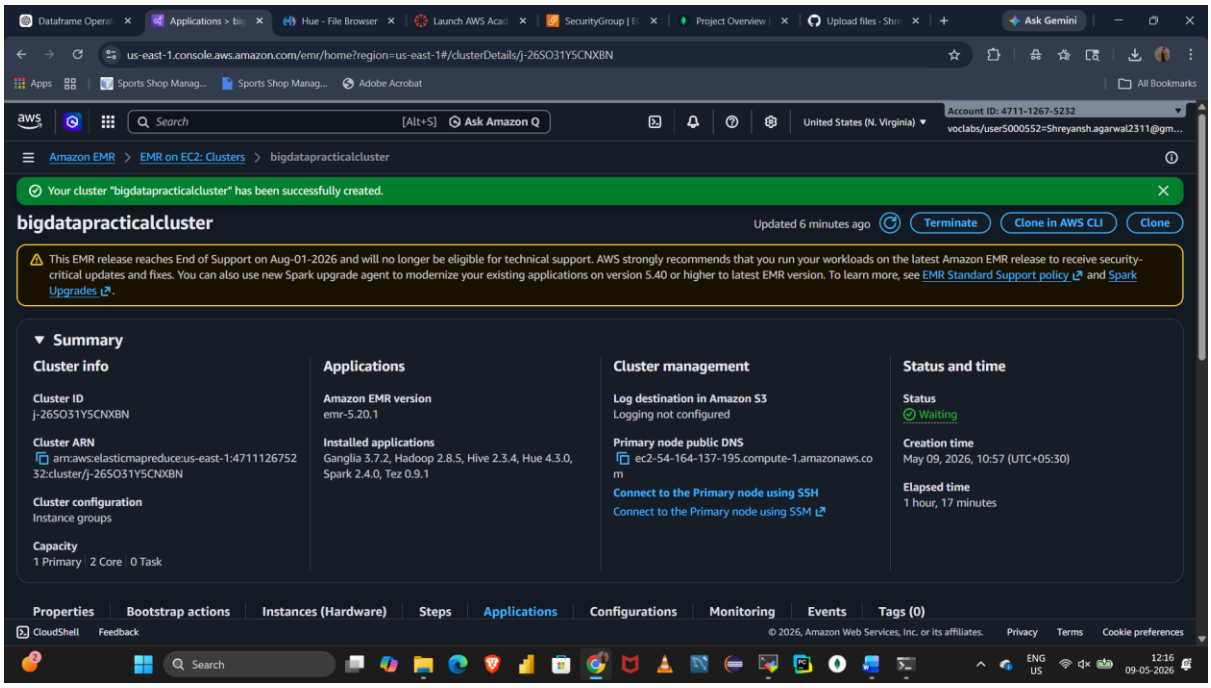
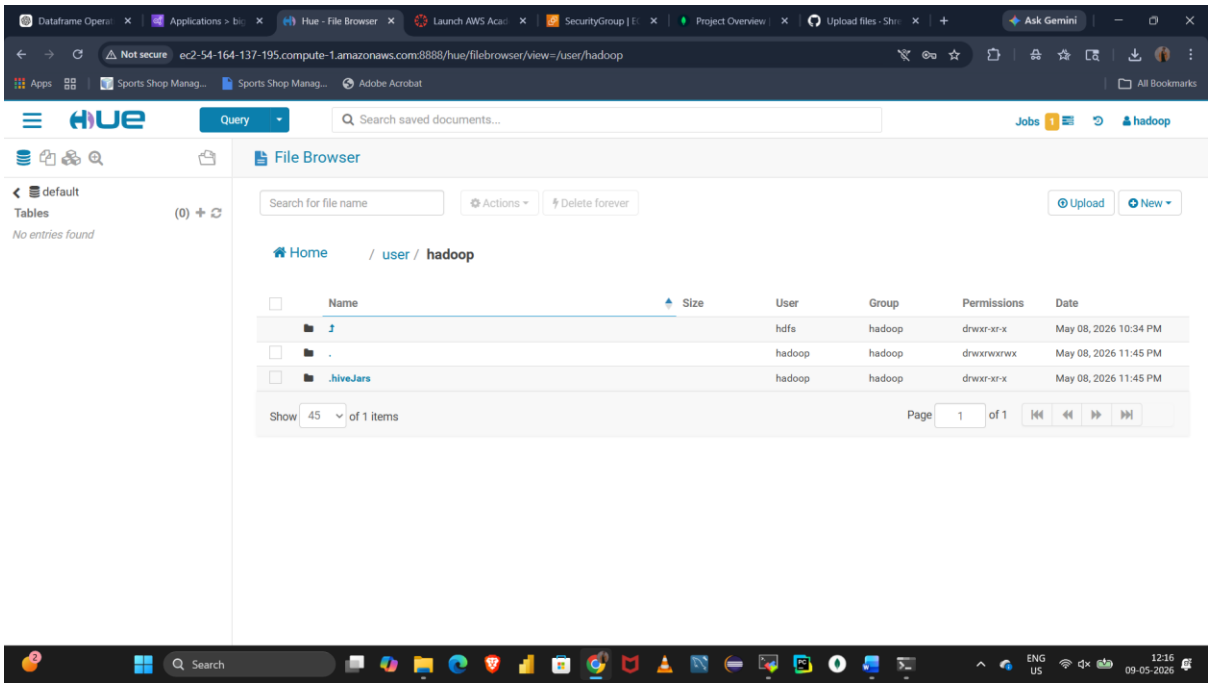
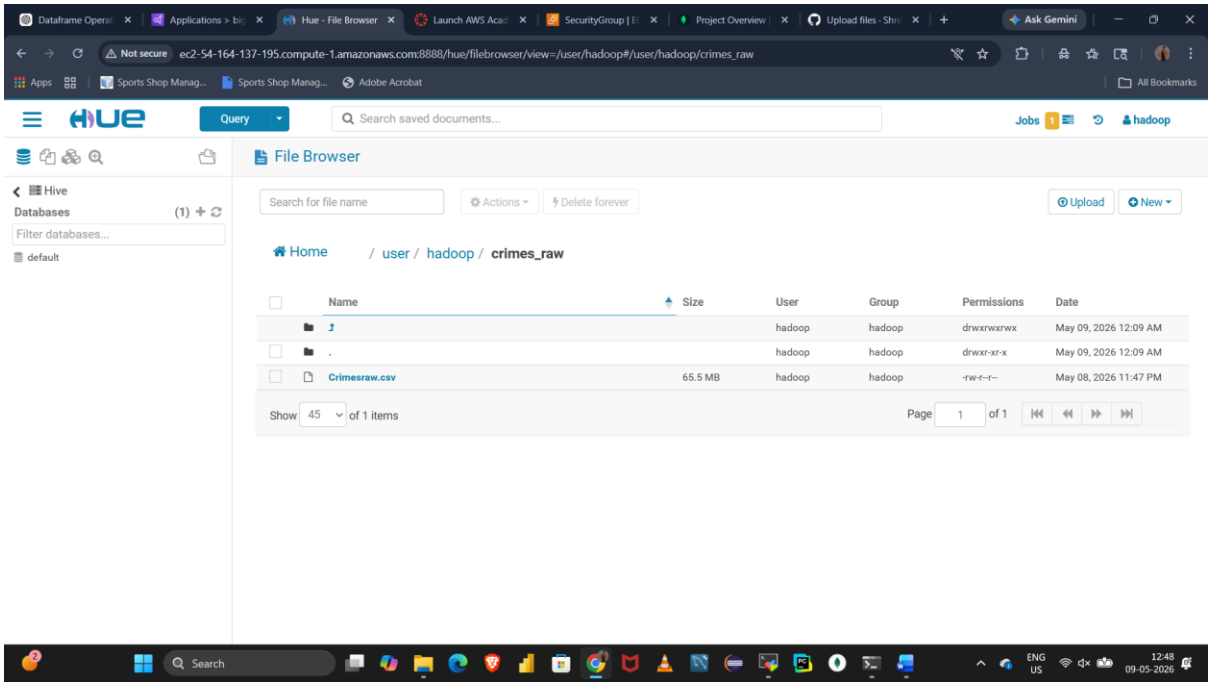
EMR cluster



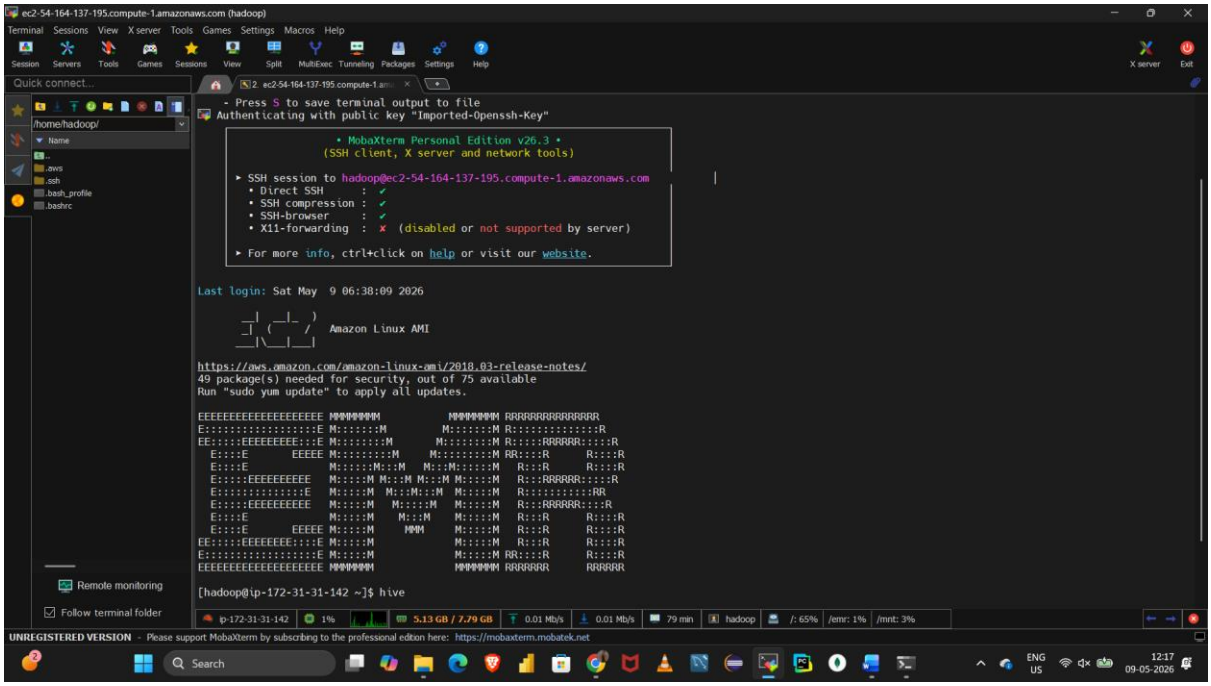
Hue



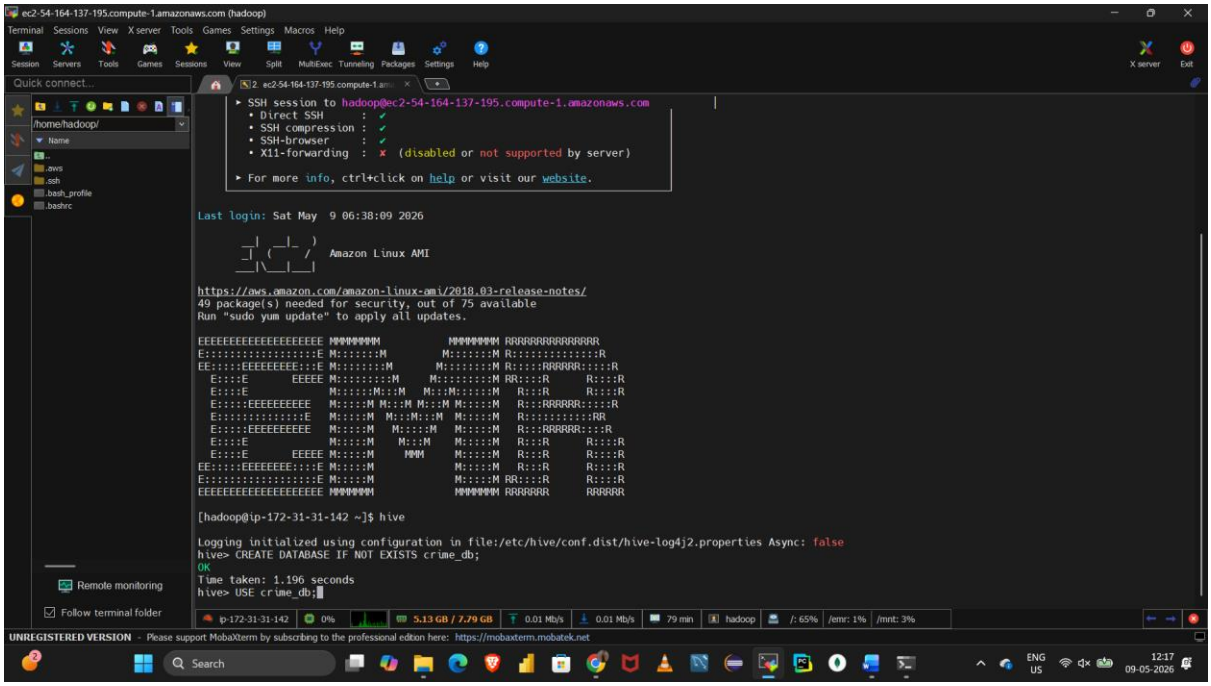
File uploading



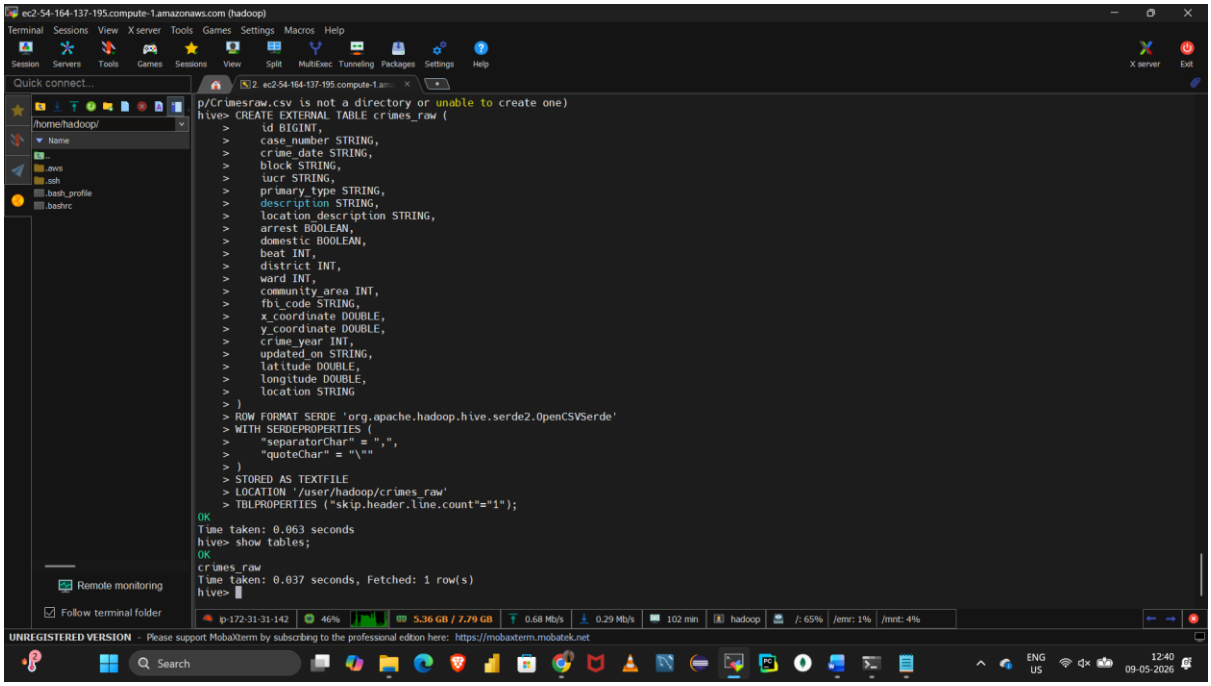
Hive in emr



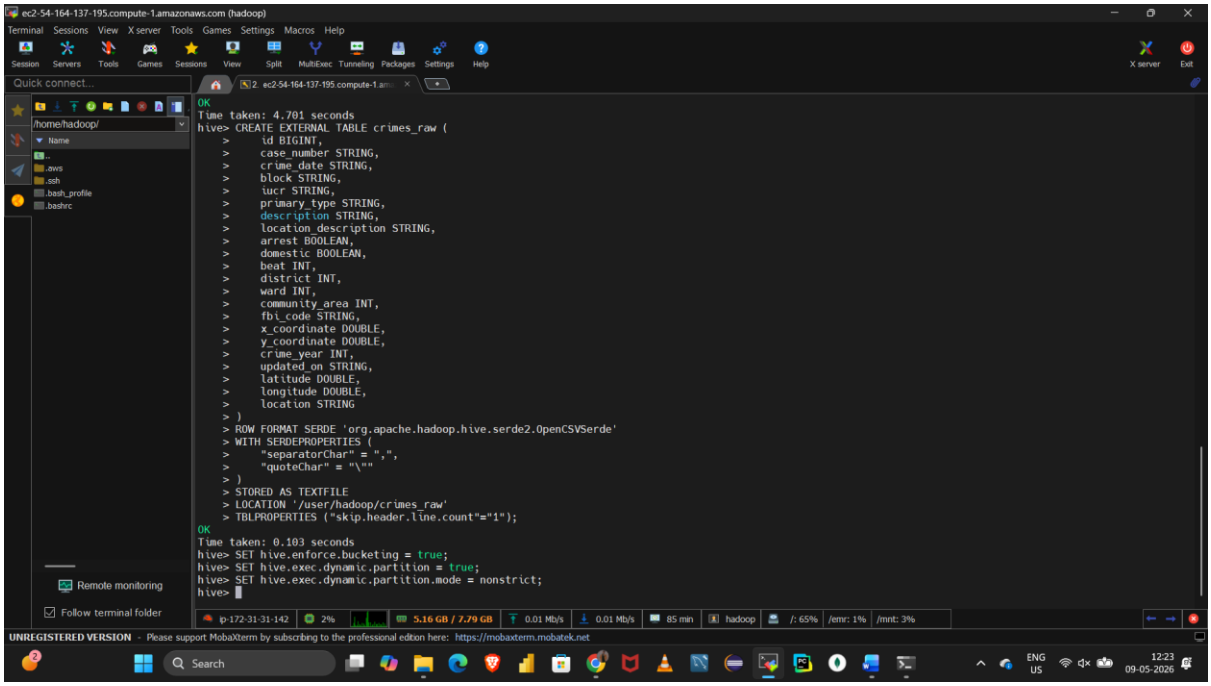
Database



Tables

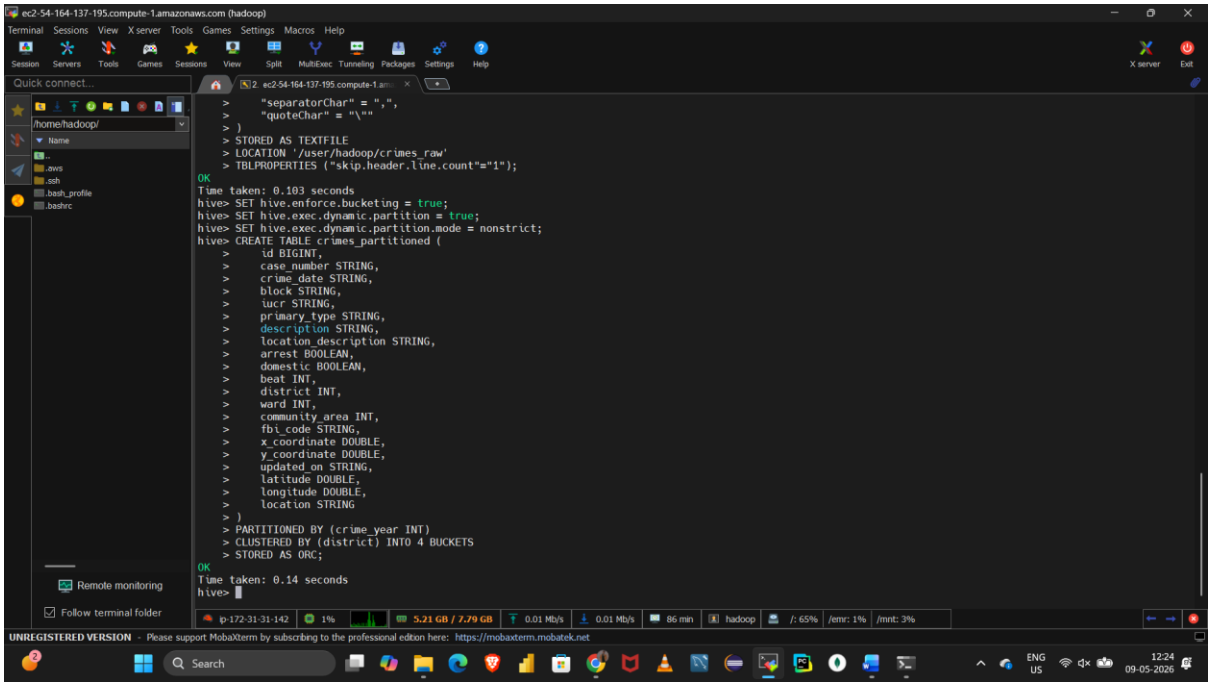


Set conditions



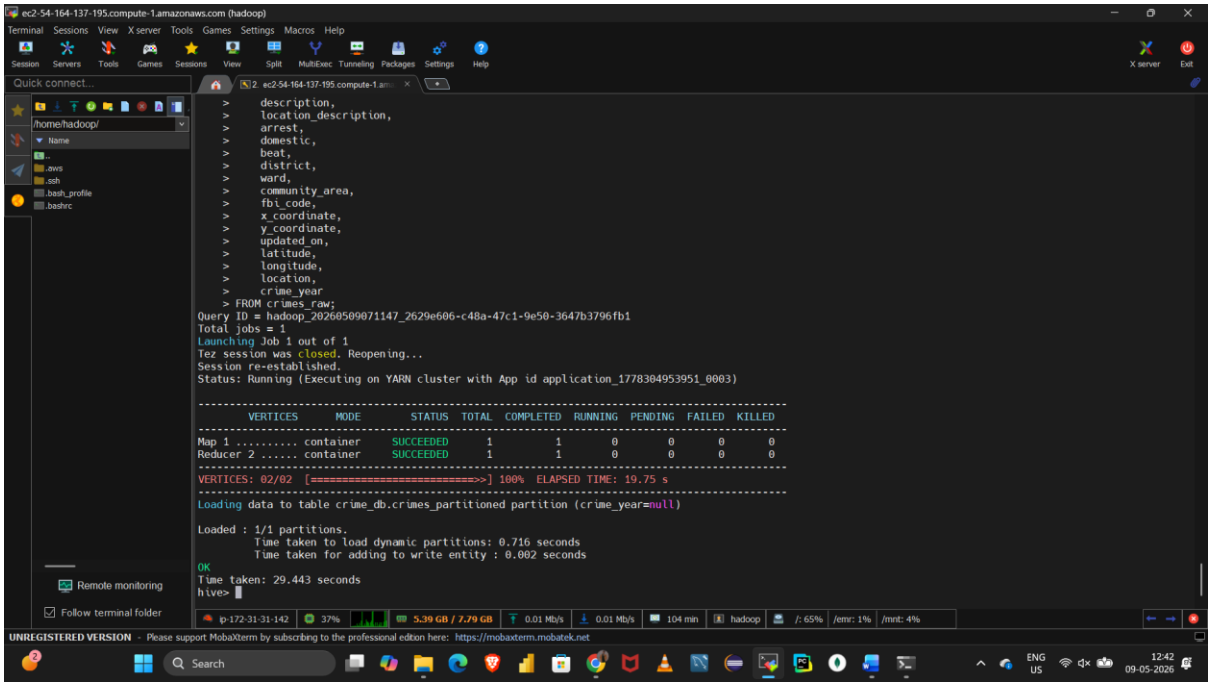
```
OK
Time taken: 4.701 seconds
hive> CREATE EXTERNAL TABLE crimes_raw (
    >     id BIGINT,
    >     case_number STRING,
    >     crime_date STRING,
    >     block STRING,
    >     iucr STRING,
    >     primary_type STRING,
    >     description STRING,
    >     location_description STRING,
    >     arrest BOOLEAN,
    >     domestic BOOLEAN,
    >     beat INT,
    >     district INT,
    >     ward INT,
    >     community_area INT,
    >     fbi_code STRING,
    >     x_coordinate DOUBLE,
    >     y_coordinate DOUBLE,
    >     crime_year INT,
    >     updated_on STRING,
    >     latitude DOUBLE,
    >     longitude DOUBLE,
    >     location STRING
    > )
    > ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.OpenCSVSerde'
    > WITH SERDEPROPERTIES (
    >     "separatorChar" = ",",
    >     "quoteChar" = "\""
    > )
    > STORED AS TEXTFILE
    > LOCATION '/user/hadoop/crimes_raw'
    > TBLPROPERTIES ("skip.header.line.count"="1");
OK
Time taken: 0.103 seconds
hive> SET hive.enforce.bucketing = true;
hive> SET hive.exec.dynamic.partition = true;
hive> SET hive.exec.dynamic.partition.mode = nonstrict;
hive>
```

Creation of partitioned table

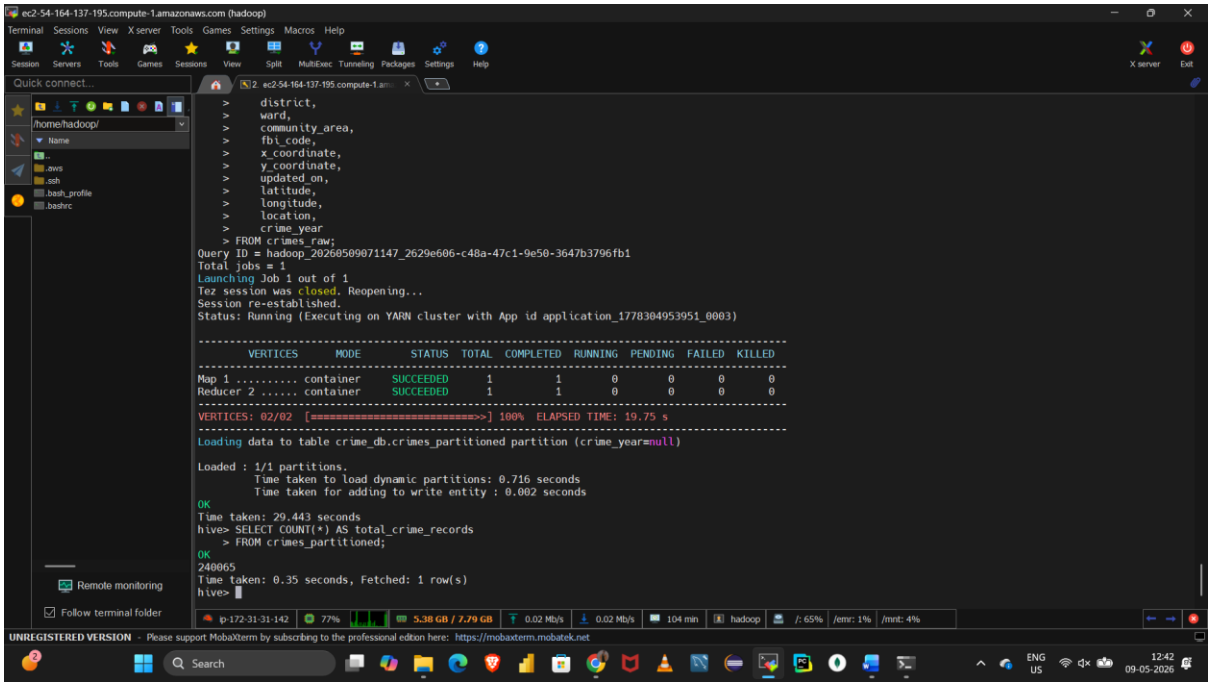


```
    >     "separatorChar" = ",",
    >     "quoteChar" = "\""
    > )
    > STORED AS TEXTFILE
    > LOCATION '/user/hadoop/crimes_raw'
    > TBLPROPERTIES ("skip.header.line.count"="1");
OK
Time taken: 0.103 seconds
hive> SET hive.enforce.bucketing = true;
hive> SET hive.exec.dynamic.partition = true;
hive> SET hive.exec.dynamic.partition.mode = nonstrict;
hive> CREATE TABLE crimes_partitioned (
    >     id BIGINT,
    >     case_number STRING,
    >     crime_date STRING,
    >     block STRING,
    >     iucr STRING,
    >     primary_type STRING,
    >     description STRING,
    >     location_description STRING,
    >     arrest BOOLEAN,
    >     domestic BOOLEAN,
    >     beat INT,
    >     district INT,
    >     ward INT,
    >     community_area INT,
    >     fbi_code STRING,
    >     x_coordinate DOUBLE,
    >     y_coordinate DOUBLE,
    >     updated_on STRING,
    >     latitude DOUBLE,
    >     longitude DOUBLE,
    >     location STRING
    > )
    > PARTITIONED BY (crime_year INT)
    > CLUSTERED BY (district) INTO 4 BUCKETS
    > STORED AS ORC;
OK
Time taken: 0.14 seconds
hive>
```

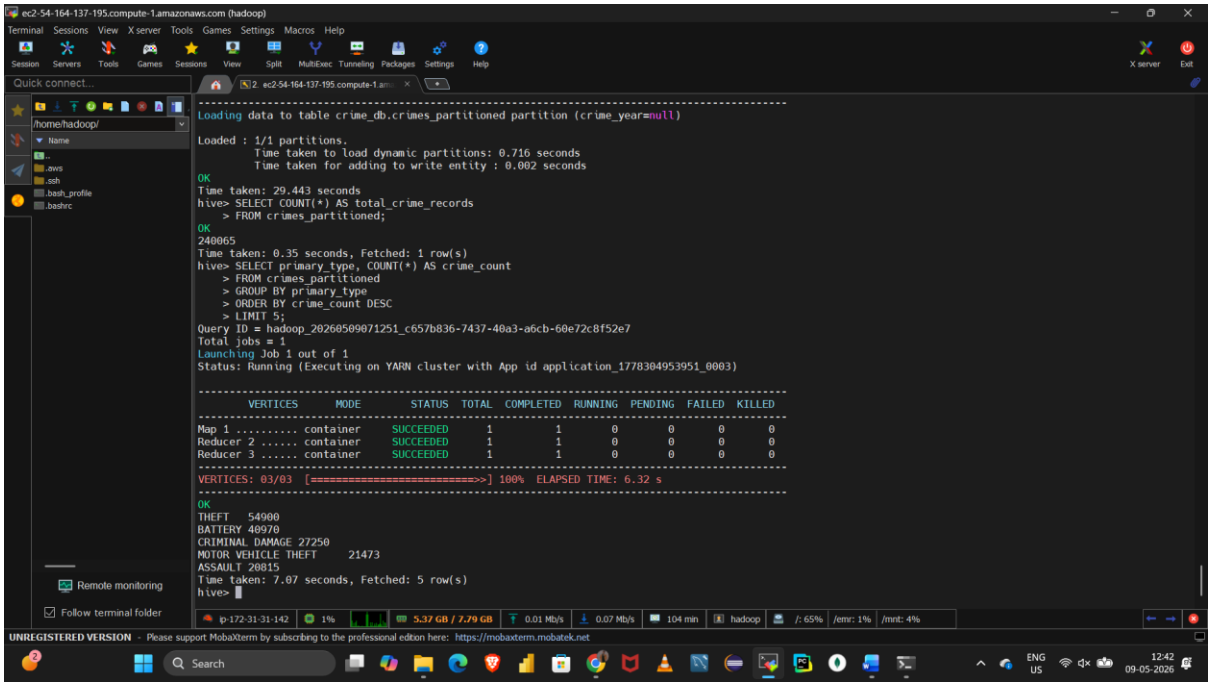
Insertion of data from crime_raw



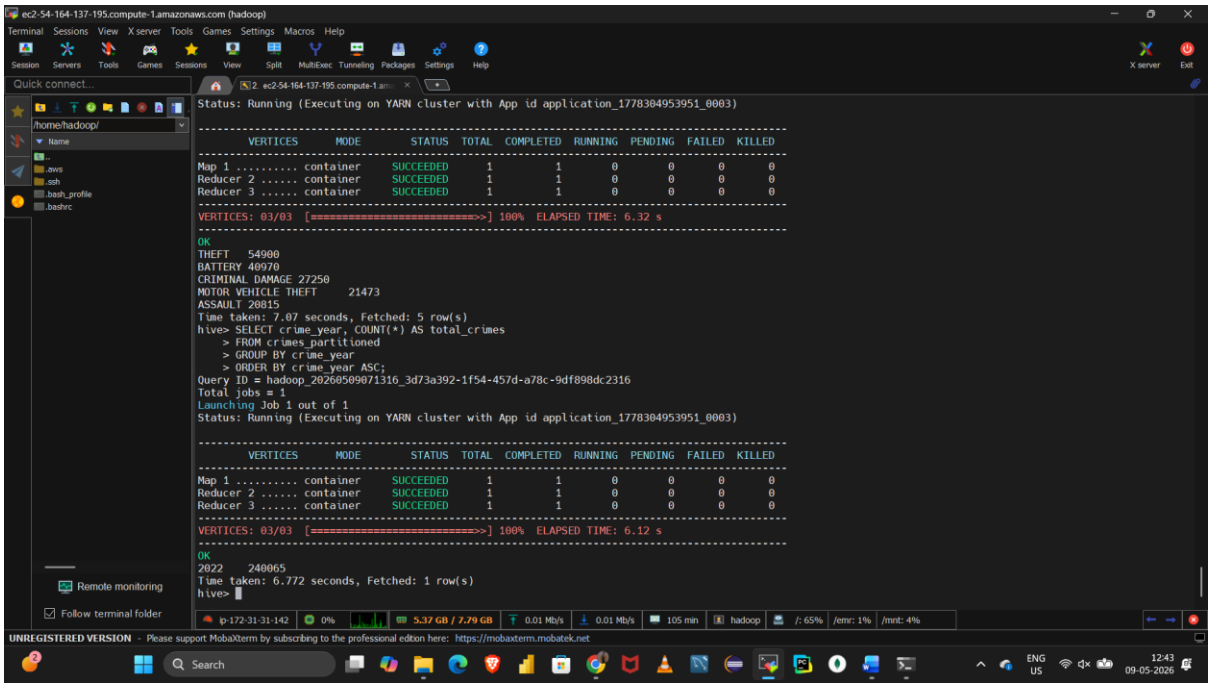
1. Count total crime records



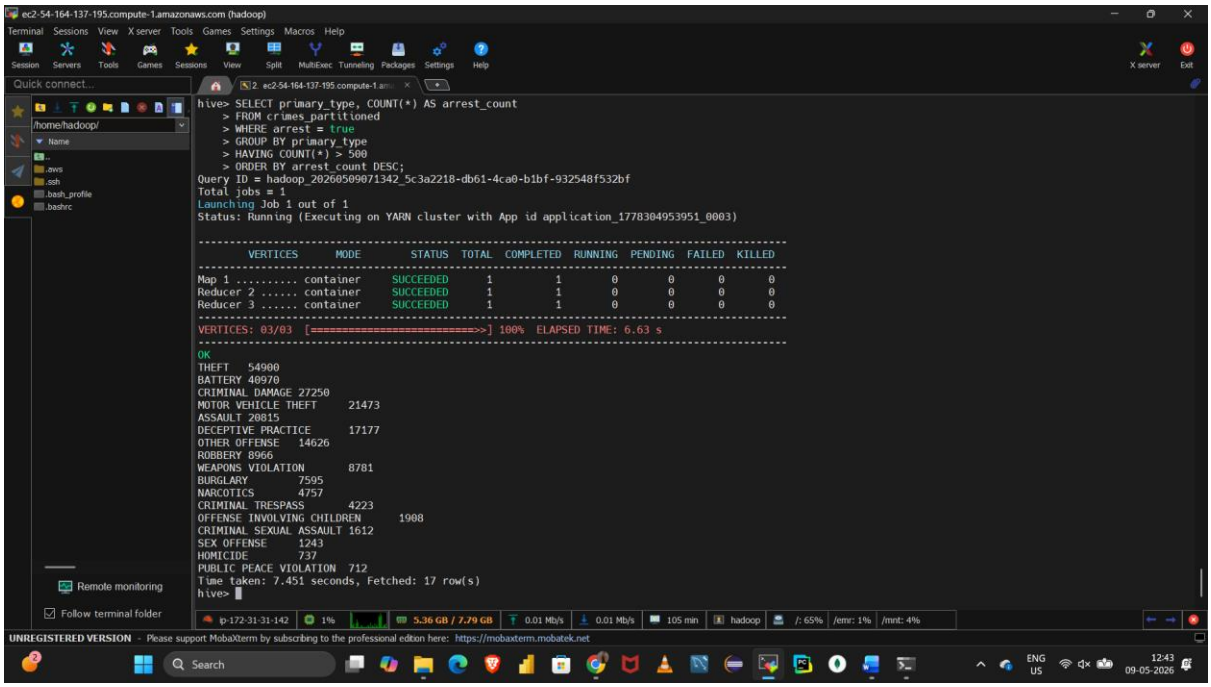
2. Top 5 most frequent crime types



3. Total crimes per year



4. Crime types with more than 500 arrests

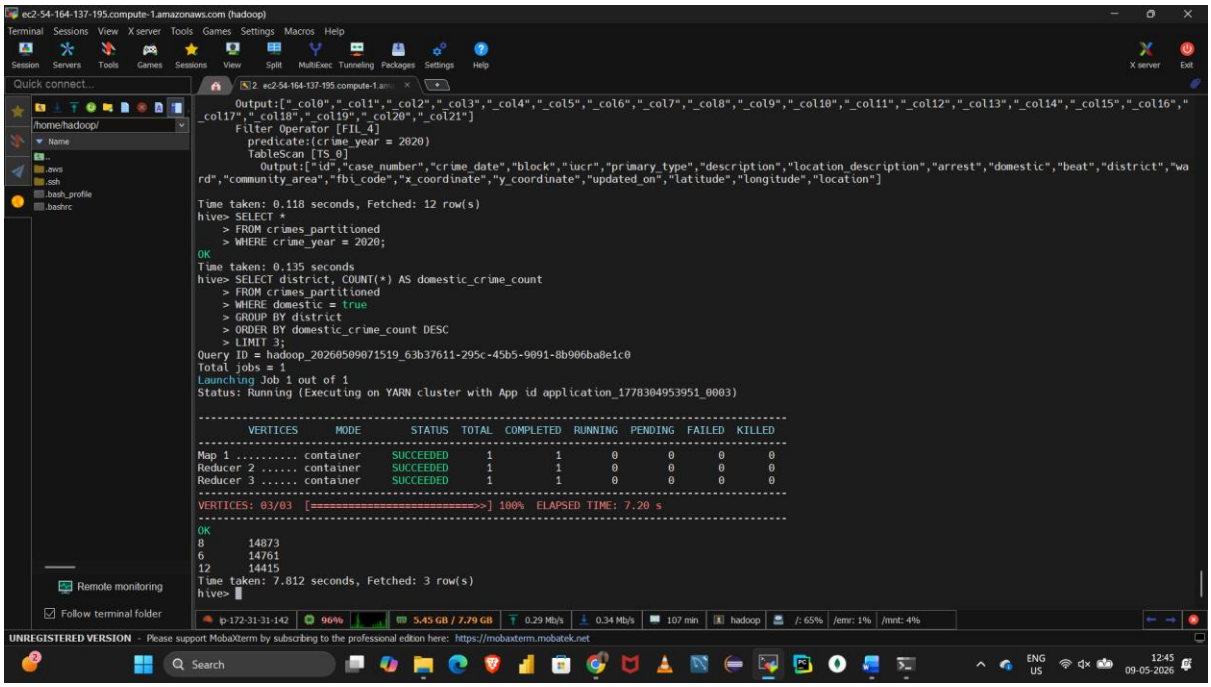


5. Retrieve all crimes for year 2020 with partition pruning

```
OK
THEFT   54900
BATTERY 40970
CRIMINAL DAMAGE 27250
MOTOR VEHICLE THEFT     21473
ASSAULT 20815
DECEPTIVE PRACTICE      17177
OTHER OFFENSE   14626
ROBBERY 8966
WEAPONS VIOLATION       8781
BURGLARY        7595
NARCOTICS       4757
CRIMINAL TRESPASS       4223
OFFENSE INVOLVING CHILDREN      1908
CRIMINAL SEXUAL ASSAULT 1612
SEX OFFENSE     1243
HOMICIDE        737
PUBLIC PEACE VIOLATION  712
Time taken: 7.451 seconds, Fetched: 17 row(s)
hive> EXPLAIN
    > SELECT *
    > FROM crimes_partitioned
    > WHERE crime_year = 2020;
OK
Plan optimized by CBO.

Stage-0
  Fetch Operator
    limit:-1
    Select Operator [SEL_2]
      Output:["_col0","_col1","_col2","_col3","_col4","_col5","_col6","_col7","_col8","_col9","_col10","_col11","_col12","_col13","_col14","_col15","_col16","
_col17","_col18","_col19","_col20","_col21"]
      Filter Operator [FIL_4]
        predicate:(crime_year = 2020)
        TableScan [TS_0]
          Output:["id","case_number","crime_date","block","iucr","primary_type","description","location_description","arrest","domestic","beat","district","wa
rd","community_area","fbi_code","x_coordinate","y_coordinate","updated_on","latitude","longitude","location"]

Time taken: 0.118 seconds, Fetched: 12 row(s)
hive>
```

```
ASSAULT 20815
DECEPTIVE PRACTICE      17177
OTHER OFFENSE   14626
ROBBERY 8966
WEAPONS VIOLATION       8781
BURGLARY        7595
NARCOTICS       4757
CRIMINAL TRESPASS       4223
OFFENSE INVOLVING CHILDREN      1908
CRIMINAL SEXUAL ASSAULT 1612
SEX OFFENSE     1243
HOMICIDE        737
PUBLIC PEACE VIOLATION  712
Time taken: 7.451 seconds, Fetched: 17 row(s)
hive> EXPLAIN
    > SELECT *
    > FROM crimes_partitioned
    > WHERE crime_year = 2020;
OK
Plan optimized by CBO.

Stage-0
  Fetch Operator
    limit:-1
    Select Operator [SEL_2]
      Output:["_col0","_col1","_col2","_col3","_col4","_col5","_col6","_col7","_col8","_col9","_col10","_col11","_col12","_col13","_col14","_col15","_col16","
_col17","_col18","_col19","_col20","_col21"]
      Filter Operator [FIL_4]
        predicate:(crime_year = 2020)
        TableScan [TS_0]
          Output:["id","case_number","crime_date","block","iucr","primary_type","description","location_description","arrest","domestic","beat","district","wa
rd","community_area","fbi_code","x_coordinate","y_coordinate","updated_on","latitude","longitude","location"]

Time taken: 0.118 seconds, Fetched: 12 row(s)
hive> SELECT *
    > FROM crimes_partitioned
    > WHERE crime_year = 2020;
OK
Time taken: 0.135 seconds
hive>
```

6. Top 3 districts with highest domestic crimes



```
      Output:["_col0","_col1","_col2","_col3","_col4","_col5","_col6","_col7","_col8","_col9","_col10","_col11","_col12","_col13","_col14","_col15","_col16","
_col17","_col18","_col19","_col20","_col21"]
      Filter Operator [FIL_4]
        predicate:(crime_year = 2020)
        TableScan [TS_0]
          Output:["id","case_number","crime_date","block","iucr","primary_type","description","location_description","arrest","domestic","beat","district","wa
rd","community_area","fbi_code","x_coordinate","y_coordinate","updated_on","latitude","longitude","location"]

Time taken: 0.118 seconds, Fetched: 12 row(s)
hive> SELECT *
    > FROM crimes_partitioned
    > WHERE crime_year = 2020;
OK
Time taken: 0.135 seconds
hive> SELECT district, COUNT(*) AS domestic_crime_count
    > FROM crimes_partitioned
    > WHERE domestic = true
    > GROUP BY district
    > ORDER BY domestic_crime_count DESC
    > LIMIT 3;
Query ID = hadoop_20260509071519_63b37611-295c-45b5-9091-8b906ba8e1c0
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1778304953951_0003)

--------------------------------------------------------------------------------
        VERTICES      MODE        STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
--------------------------------------------------------------------------------
Map 1 ......... container     SUCCEEDED      1          1        0        0       0       0
Reducer 2 ...... container     SUCCEEDED      1          1        0        0       0       0
Reducer 3 ...... container     SUCCEEDED      1          1        0        0       0       0
--------------------------------------------------------------------------------
VERTICES: 03/03  [==========================>>] 100%  ELAPSED TIME: 7.20 s
--------------------------------------------------------------------------------
OK
8       14873
6       14761
12      14415
Time taken: 7.812 seconds, Fetched: 3 row(s)
hive>
```